

ARNOLD ABRAHAM

SOFTWARE ENGINEER

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Location: Erlangen, Germany

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Experience

Software Engineer – Working Student

Jul 2023 – Jun 2025

Siemens Logistics (Vanderlande), Nuremberg, Germany

- Led the end-to-end migration of two Java applications from Oracle to PostgreSQL, ensuring database compatibility, data integrity, and application stability while reducing database licensing costs.
- Developed backend features and optimized database interfaces to enhance performance and maintainability.
- Designed and implemented automated installation workflows for JRE, PostgreSQL, and environment configuration, reducing setup effort by 70%.
- Built a Python data pipeline for predictive maintenance, integrating ML models with real-time industrial monitoring.
- Implemented an async-io-based Python data acquisition system for concurrent real-time monitoring of industrial drives.
- Collaborated with cross-functional engineering teams to integrate predictive monitoring into production workflows.

Product Engineer

Sept 2021 – Sept 2022

IBS Software, Kochi, India

- Developed Java/Spring-based backend components for an airline crew scheduling platform used by LATAM and Qantas.
- Implemented and maintained complex scheduling and compliance business logic, including bug fixes and enhancements driven by production and QA feedback.
- Created, maintained, and troubleshoot SQL scripts supporting backend functionality and ensuring correct system behavior.
- Contributed across the full software development lifecycle, including design, testing (JUnit), code reviews, and production issue resolution.

Education

MSc Data Science

Oct 2022 – Jul 2025

University of Erlangen–Nuremberg, Germany

- Major in Artificial Intelligence and Machine Learning.
- Thesis: Predictive Maintenance for Electrical Panels: Hotspot Forecasting and Anomaly Detection Using Thermal Data (Siemens).

B.Tech Computer Science and Engineering

Aug 2017 – Jul 2021

APJ Abdul Kalam Technological University, India

Technical Skills

Programming Languages: Java, Python, SQL, JavaScript

Frameworks: Spring Boot, Spring, Hibernate (JPA), React, JUnit

Tools & Platforms: Git, GitLab, Bitbucket, Docker, AWS, CI/CD, JIRA, Confluence, Swagger UI

Databases: PostgreSQL, Oracle, InfluxDB

Core Competencies

Backend Development ▪ Distributed Systems ▪ REST APIs ▪ Object-Oriented Design ▪ Data Structures & Algorithms
▪ SDLC ▪ Unit Testing ▪ CI/CD ▪ ML & Time-Series Forecasting

Languages

English (C1 – Fluent) ▪ German (A1 – Basic)